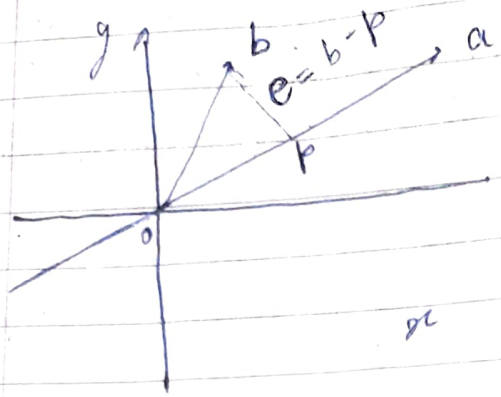
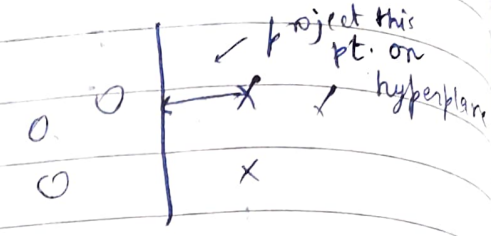
Projection



map vector b onto vector a



error = e = b - p

$p = Xa$ (since p is a portion of a)

e is perpendicular to a
or
orthogonal

$a^T e = 0$ - dot product is 0

$a^T(b - Xa) = 0$

$Xa^Ta = a^Tb$

$X = \dfrac{a^Tb}{a^Ta}$

$a^T \rightarrow$ vector, a = scalar
dot product = scaler
which is a number
∴ division is OK.

$$\boxed{p = a\left(\frac{a^Tb}{a^Ta}\right)}$$

p = Project^n matrix (PM) b

where

$PM = \dfrac{aa^T}{a^Ta}$

Projection matrix in 1D

why not cancel?

$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\quad aa^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$

↳ matrix

$a^Ta \Rightarrow$ scalar

$PM = A(A^TA)^{-1}A^T$

let $b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

MATRIX MULTIPLICATION

$(m \times n)(n \times p) \rightarrow (m \times p)$

$\therefore (2 \times 1)(1 \times 3) \rightarrow \checkmark$

$(2 \times \underbrace{1)\ (2}_{\text{not same}} \times 2) \rightarrow \times$

$\therefore (2 \times 1)(1 \times 2) \rightarrow (2 \times 2)$

$a = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \Rightarrow$ horizontal line along the X-axis

find P, i.e. project b onto a

$a^T = \begin{bmatrix} 4 & 0 \end{bmatrix}$

$a^Tb = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$a^Ta = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

$= 16$

$(2 \times 1) \times (1 \times 2) \longrightarrow (2 \times 2)$

$aa^T = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \begin{bmatrix} 4 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$

$\frac{aa^T}{a^Ta} \Rightarrow \frac{1}{16} \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$(2 \times 2) \times (2 \times 1) \rightarrow (2 \times 1)$